

1. Manufacturing Production Analysis

A manufacturing company records daily production, machine operating hours, temperature, energy consumption, and defect count. Import and explore the dataset using Pandas. Develop visualizations to analyze production trends over time, relationships between machine parameters and defects, and distributions of production values. Use a scatter matrix, lag plot, and autocorrelation plot to identify hidden production patterns and provide recommendations for improving efficiency.

SOLUTION:

Objective

To analyze daily production, machine operating hours, temperature, energy consumption, and defect count using Pandas. The analysis identifies production trends, relationships between machine parameters and defects, distributions, and hidden time-based patterns.

Dataset

Assume the dataset is stored as:

manufacturing_production.csv

Columns:

- Date
- Production
- Operating_Hours
- Temperature
- Energy_Consumption
- Defect_Count

Complete Python Code

```
import pandas as pd  
  
import matplotlib.pyplot as plt  
  
import seaborn as sns
```

```
from pandas.plotting import scatter_matrix, lag_plot

df = pd.read_csv("manufacturing_production.csv")

print("First 5 rows:")

print(df.head())

print("\nDataset Shape:")

print(df.shape)

print("\nColumn Names:")

print(df.columns)

print("\nDataset Information:")

print(df.info())

print("\nStatistical Summary:")

print(df.describe())

print("\nMissing Values:")

print(df.isnull().sum())

df["Date"] = pd.to_datetime(df["Date"])

df = df.sort_values("Date")

print("\nSorted Dataset:")

print(df.head())

plt.figure(figsize=(10, 5))

plt.plot(
    df["Date"],
    df["Production"]
)

plt.title("Daily Production Trend")

plt.xlabel("Date")

plt.ylabel("Production")

plt.xticks(rotation=45)

plt.tight_layout()

plt.show()

plt.figure(figsize=(7, 5))

sns.scatterplot(
    data=df,
    x="Operating_Hours",
```

```

        y="Production"
    )
    plt.title("Operating Hours vs Production")
    plt.xlabel("Operating Hours")
    plt.ylabel("Production")
    plt.show()
    plt.figure(figsize=(7, 5))
    sns.scatterplot(
        data=df,
        x="Temperature",
        y="Defect_Count"
    )
    plt.title("Temperature vs Defect Count")
    plt.xlabel("Temperature")
    plt.ylabel("Defect Count")
    plt.show()
    plt.figure(figsize=(7, 5))
    sns.scatterplot(
        data=df,
        x="Energy_Consumption",
        y="Production"
    )
    plt.title("Energy Consumption vs Production")
    plt.xlabel("Energy Consumption")
    plt.ylabel("Production")
    plt.show()
    plt.figure(figsize=(7, 5))
    sns.histplot(
        df["Production"],
        kde=True
    )
    plt.title("Production Distribution")
    plt.xlabel("Production")

```

```
plt.ylabel("Frequency")

plt.show()

plt.figure(figsize=(7, 5))

sns.histplot(
    df["Defect_Count"],
    kde=True
)

plt.title("Defect Count Distribution")

plt.xlabel("Defect Count")

plt.ylabel("Frequency")

plt.show()

scatter_matrix(
    df[
        [
            "Production",
            "Operating_Hours",
            "Temperature",
            "Energy_Consumption",
            "Defect_Count"
        ]
    ],
    figsize=(12, 10),
    diagonal="hist"
)

plt.suptitle("Scatter Matrix of Production Variables")

plt.show()

plt.figure(figsize=(7, 5))

lag_plot(
    df["Production"],
    lag=1
)

plt.title("Lag Plot of Daily Production")

plt.show()
```

```

plt.figure(figsize=(10, 5))
pd.plotting.autocorrelation_plot(
    df["Production"]
)
plt.title("Autocorrelation of Daily Production")
plt.show()
correlation = df[
    [
        "Production",
        "Operating_Hours",
        "Temperature",
        "Energy_Consumption",
        "Defect_Count"
    ]
].corr()
print("\nCorrelation Matrix:")
print(correlation)
print("\nCorrelation with Production:")
print(
    correlation["Production"]
    .sort_values(ascending=False)
)
print("\nCorrelation with Defect Count:")
print(
    correlation["Defect_Count"]
    .sort_values(ascending=False)
)
Q1 = df["Production"].quantile(0.25)
Q3 = df["Production"].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
outliers = df[

```

```

(df["Production"] < lower) |
(df["Production"] > upper)
]
print("\nProduction Outliers:")
print(
    outliers[
        ["Date", "Production"]
    ]
)
print("\nNumber of Production Outliers:")
print(len(outliers))

```

Sample Output

The exact values depend on the manufacturing dataset used.

Dataset Exploration

First 5 rows:

	Date	Production	Operating_Hours	Temperature	Energy_Consumption	Defect_Count
0	2024-01-01	5200	8.2	28.5	420.5	12
1	2024-01-02	5350	8.5	29.1	435.2	10
2	2024-01-03	5500	8.8	30.2	450.7	9
3	2024-01-04	5100	7.9	31.5	410.3	15
4	2024-01-05	5700	9.1	29.8	465.8	8

Dataset Shape

Dataset Shape:

(365, 6)

Statistical Summary

	Production	Operating_Hours	Temperature
count	365.00	365.00	365.00
mean	5480.25	8.42	30.15
std	520.42	0.85	3.21
min	4100.00	6.20	22.10

max 6900.00 10.20 38.50

Missing Values

Date 0

Production 0

Operating_Hours 2

Temperature 1

Energy_Consumption 3

Defect_Count 0

Correlation Output

 Production Operating_Hours Temperature Energy_Consumption
Defect_Count

Production	1.00	0.82	0.18	0.76	-0.48
Operating_Hours	0.82	1.00	0.12	0.73	-0.39
Temperature	0.18	0.12	1.00	0.15	0.64
Energy_Consumption	0.76	0.73	0.15	1.00	-0.31
Defect_Count	-0.48	-0.39	0.64	-0.31	1.00

Interpretation

- Operating hours and production: strong positive relationship.
- Energy consumption and production: positive relationship.
- Temperature and defects: positive relationship.
- Production and defects: moderate negative relationship.

Correlation indicates association, not necessarily causation.

Production Outliers

Production Outliers:

	Date	Production
45	2024-02-15	4100
178	2024-06-27	6900
291	2024-10-18	4050

Number of Production Outliers:

3

These values should be investigated to determine whether they resulted from machine downtime, maintenance, unusual demand, or data-entry errors.

Scatter Matrix

The scatter matrix helps compare all numerical variables simultaneously. It can reveal:

- Positive relationships between operating hours and production.
- Positive relationships between energy consumption and production.
- Relationships between temperature and defects.
- Possible clusters and unusual observations.

Lag Plot

The lag plot compares today's production with the previous day's production.

If the points form a clear pattern instead of being randomly scattered, daily production has dependence on previous production values.

Autocorrelation Plot

The autocorrelation plot identifies whether production values are related to their previous values at different time lags.

High autocorrelation at small lags indicates that production has time-dependent behavior, meaning yesterday's production can provide information about today's production.

Major Findings

1. Machine operating hours strongly influence production. More operating hours are generally associated with higher production.
2. Energy consumption increases with production, indicating that higher output requires greater energy usage.
3. Higher temperatures are associated with more defects, suggesting that excessive operating temperature may affect product quality.
4. Daily production shows time dependence, as revealed by the lag and autocorrelation analysis.
5. Production outliers indicate unusual operating conditions and should be investigated for machine downtime, maintenance, or abnormal production demand.

Recommendations

- Optimize machine operating hours to increase production without excessive machine usage.

- Monitor energy consumption per unit of production to improve energy efficiency.
- Maintain suitable machine temperature levels to reduce defects.
- Use previous production data for production forecasting and planning.
- Investigate production outliers and establish automatic alerts for abnormal machine behavior.

Conclusion

The analysis shows that operating hours and energy consumption are strongly associated with production, while temperature has an important relationship with defects. The scatter matrix, lag plot, and autocorrelation plot help reveal relationships and hidden time-based patterns that may not be visible from basic statistics alone. These findings can help the manufacturing company improve production efficiency, energy usage, and product quality.

2. E-Commerce Sales Pattern Analysis

An e-commerce company has historical data containing date, product category, quantity sold, revenue, discount, and customer rating. Analyze the evolution of sales over time and identify relationships between discount, quantity, and revenue. Use appropriate bar charts, line charts, distribution plots, frequency plots, and scatter matrices to identify important business patterns.

SOLUTION:

Objective

To analyze historical e-commerce data and identify sales trends over time, relationships between discount, quantity sold, and revenue, and important business patterns using different statistical visualizations.

Dataset

Assume the dataset is:

ecommerce_sales.csv

Columns:

- Date
- Product_Category
- Quantity_Sold
- Revenue
- Discount
- Customer_Rating

Complete Python Code

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

df = pd.read_csv("ecommerce_sales.csv")

print("First 5 Rows:")

print(df.head())

print("\nDataset Shape:")

print(df.shape)

print("\nColumn Names:")

print(df.columns)

print("\nDataset Information:")

print(df.info())

print("\nMissing Values:")

print(df.isnull().sum())

print("\nStatistical Summary:")

print(df.describe())

df["Date"] = pd.to_datetime(df["Date"])

df = df.sort_values("Date")

df["Month"] = df["Date"].dt.to_period("M")

monthly_sales = df.groupby("Month")["Revenue"].sum()

print("\nMonthly Revenue:")

print(monthly_sales)

category_sales = df.groupby(

    "Product_Category"

)["Revenue"].sum().sort_values(ascending=False)

print("\nRevenue by Product Category:")

print(category_sales)

category_quantity = df.groupby(

    "Product_Category"

)["Quantity_Sold"].sum().sort_values(ascending=False)

print("\nQuantity Sold by Category:")

print(category_quantity)
```

```

correlation = df[
    "Quantity_Sold",
    "Revenue",
    "Discount",
    "Customer_Rating"
].corr()
print("\nCorrelation Matrix:")
print(correlation)
plt.figure(figsize=(8, 5))
category_sales.plot(kind="bar")
plt.title("Revenue by Product Category")
plt.xlabel("Product Category")
plt.ylabel("Revenue")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
plt.figure(figsize=(10, 5))
plt.plot(
    monthly_sales.index.astype(str),
    monthly_sales.values,
    marker="o"
)
plt.title("Monthly Revenue Trend")
plt.xlabel("Month")
plt.ylabel("Revenue")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
plt.figure(figsize=(8, 5))
sns.histplot(
    df["Revenue"],

```

```
kde=True
)
plt.title("Revenue Distribution")
plt.xlabel("Revenue")
plt.ylabel("Frequency")
plt.show()
plt.figure(figsize=(8, 5))
sns.histplot(
    df["Quantity_Sold"],
    bins=15
)
plt.title("Frequency Distribution of Quantity Sold")
plt.xlabel("Quantity Sold")
plt.ylabel("Frequency")
plt.show()
plt.figure(figsize=(8, 5))
sns.scatterplot(
    data=df,
    x="Discount",
    y="Revenue"
)
plt.title("Discount vs Revenue")
plt.xlabel("Discount (%)")
plt.ylabel("Revenue")
plt.show()
plt.figure(figsize=(8, 5))
sns.scatterplot(
    data=df,
    x="Quantity_Sold",
    y="Revenue"
)
plt.title("Quantity Sold vs Revenue")
plt.xlabel("Quantity Sold")
```

```

plt.ylabel("Revenue")
plt.show()

from pandas.plotting import scatter_matrix

scatter_matrix(
    df[
        [
            "Quantity_Sold",
            "Revenue",
            "Discount",
            "Customer_Rating"
        ]
    ],
    figsize=(12, 10),
    diagonal="hist"
)

plt.suptitle("Scatter Matrix of E-Commerce Variables")
plt.show()

print("\nHighest Revenue Category:")
print(category_sales.idxmax())

print("\nHighest Revenue Month:")
print(monthly_sales.idxmax())

print("\nAverage Discount:")
print(df["Discount"].mean())

print("\nAverage Customer Rating:")
print(df["Customer_Rating"].mean())

```

Sample Output

The values below are sample outputs. Actual results depend on the dataset used.

Dataset Exploration

First 5 Rows:

	Date	Product_Category	Quantity_Sold	Revenue	Discount	Customer_Rating
0	2024-01-05	Electronics	12	4800	10	4.5

1	2024-01-08	Clothing	8	2400	15	4.1
2	2024-01-12	Electronics	15	6200	5	4.7
3	2024-01-18	Furniture	5	3500	20	3.9
4	2024-01-22	Books	20	1800	10	4.3

Dataset Shape

Dataset Shape:

(500, 6)

Missing Values

Date 0

Product_Category 0

Quantity_Sold 0

Revenue 0

Discount 5

Customer_Rating 3

dtype: int64

Statistical Summary

	Quantity_Sold	Revenue	Discount	Customer_Rating
count	500.00	500.00	495.00	497.00
mean	15.42	4250.60	12.40	4.18
std	8.21	2350.45	7.20	0.52
min	1.00	350.00	0.00	2.50
max	45.00	12500.00	30.00	5.00

Monthly Revenue Output

Monthly Revenue:

Month

2024-01 125000

2024-02 138000

2024-03 142000

2024-04 151000

2024-05	165000
2024-06	158000
2024-07	172000
2024-08	181000
2024-09	176000
2024-10	195000
2024-11	245000
2024-12	285000

Observation: Sales generally increase throughout the year, with particularly high revenue during November and December.

Revenue by Product Category

Revenue by Product Category:

Electronics	820000
Clothing	650000
Furniture	480000
Beauty	390000
Books	290000

Observation: Electronics generates the highest revenue.

Correlation Matrix

	Quantity_Sold	Revenue	Discount	Customer_Rating
--	---------------	---------	----------	-----------------

Quantity_Sold	1.00	0.78	0.21	0.18
Revenue	0.78	1.00	0.15	0.25
Discount	0.21	0.15	1.00	0.05
Customer_Rating	0.18	0.25	0.05	1.00

Interpretation

- Quantity Sold and Revenue = 0.78 → strong positive relationship.
- Discount and Revenue = 0.15 → weak positive relationship.
- Customer Rating and Revenue = 0.25 → weak positive relationship.

So, quantity sold has the strongest relationship with revenue in this example.

Important Business Patterns

1. Sales Trend

Revenue increases over time, with the highest sales occurring toward the end of the year.

2. Product Category

Electronics contributes the highest revenue and should receive sufficient inventory and marketing attention.

3. Quantity and Revenue

There is a strong positive relationship between quantity sold and revenue. Increasing the number of products sold generally increases revenue.

4. Discount and Revenue

Discount has only a weak relationship with revenue. Therefore, simply providing larger discounts does not necessarily guarantee higher revenue.

5. Customer Rating

Higher-rated products show a slight tendency toward higher revenue, suggesting that customer satisfaction may contribute to sales.

Final Recommendations

1. Increase inventory before peak periods, especially November and December.
2. Focus marketing efforts on high-revenue categories such as Electronics.
3. Use discounts strategically instead of offering large discounts on all products.
4. Improve product quality and customer experience to maintain high customer ratings.
5. Monitor quantity sold and revenue together to identify products with strong sales potential.

Conclusion

The analysis shows that sales evolve over time and increase during peak periods, while quantity sold has the strongest relationship with revenue. Bar charts, line charts, distribution plots, frequency plots, and scatter matrices help identify these patterns and support better decisions regarding inventory, pricing, discounts, and marketing strategies.